

Method Overriding

Method Overriding is a **feature of runtime polymorphism** where a subclass provides its own implementation of a method that is already defined in its superclass.

Basic Method Overriding

```
class Animal {  
    void sound() {  
        System.out.println("Animal makes a sound");  
    }  
}
```

```
class Dog extends Animal {  
    @Override  
    void sound() {  
        System.out.println("Dog barks");  
    }  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        Dog d = new Dog();  
        d.sound();  
    }  
}
```

Dog barks

```
class Animal {  
    void sound() {  
        System.out.println("Animal makes a sound");  
    }  
}
```

```
class Dog extends Animal {  
    @Override  
    void sounds() { // Error  
        System.out.println("Dog barks");  
    }  
}
```

Parent Object

```
class Animal {  
    void sound() {  
        System.out.println("Animal Sound");  
    }  
}
```

```
class Dog extends Animal {
    @Override
    void sound() {
        System.out.println("Dog Barks");
    }
}

public class Main {
    public static void main(String[] args) {
        Animal obj = new Animal();
        obj.sound();
    }
}

Animal Sound
```

Runtime Polymorphism

```
class Animal {
    void sound() {
        System.out.println("Animal Sound");
    }
}

class Dog extends Animal {
    @Override
    void sound() {
        System.out.println("Dog Barks");
    }
}

public class Main {
    public static void main(String[] args) {
        Animal obj = new Dog();
        obj.sound();
    }
}

Dog Barks
```

Multiple Classes

```
class Shape {
    void draw() {
        System.out.println("Drawing Shape");
    }
}
```

```
class Circle extends Shape {
    @Override
    void draw() {
        System.out.println("Drawing Circle");
    }
}

class Rectangle extends Shape {
    @Override
    void draw() {
        System.out.println("Drawing Rectangle");
    }
}

public class Main {
    public static void main(String[] args) {
        Shape s1 = new Circle();
        Shape s2 = new Rectangle();
        s1.draw();
        s2.draw();
    }
}
```

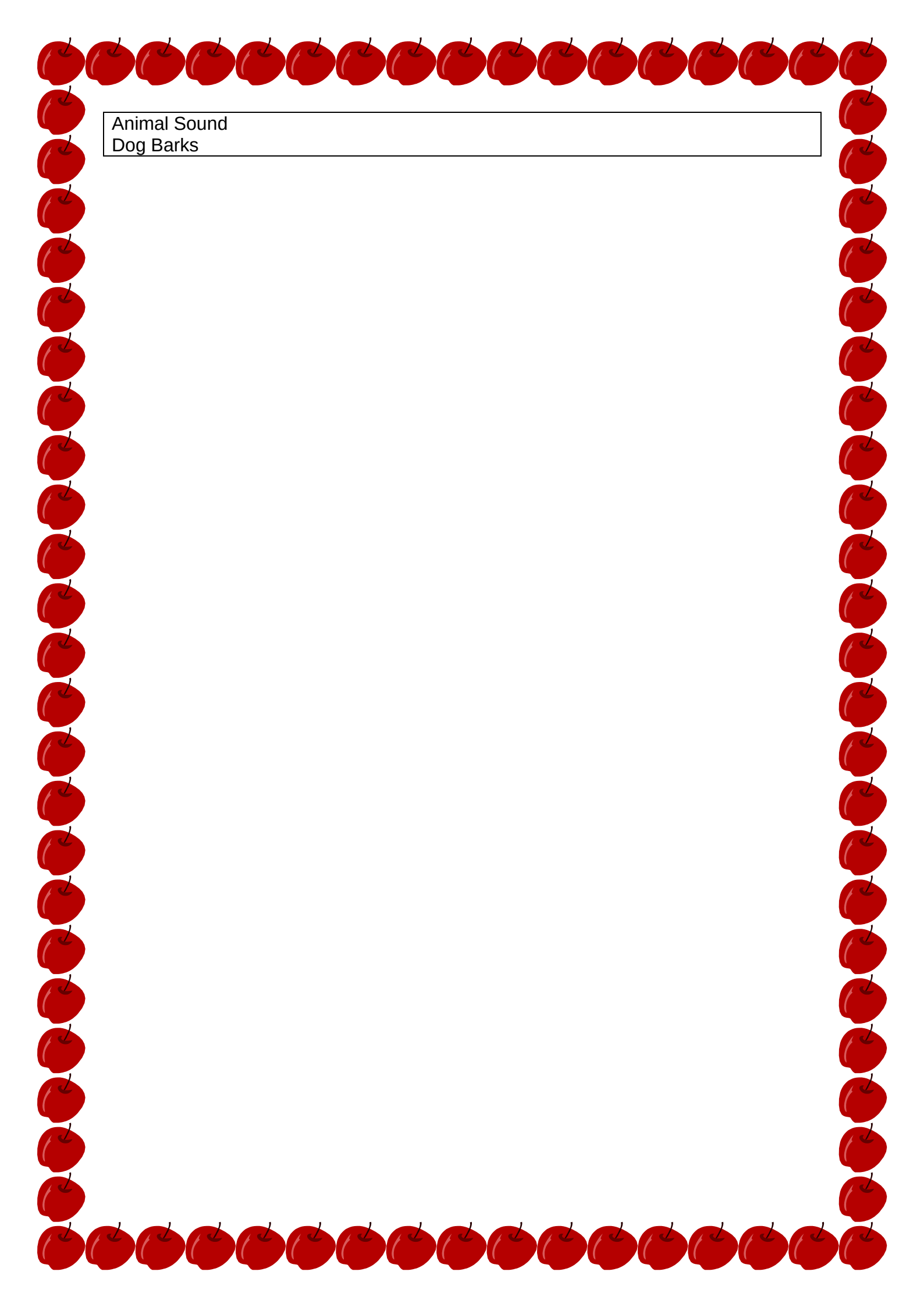
Drawing Circle
Drawing Rectangle

Using super

```
class Animal {
    void sound() {
        System.out.println("Animal Sound");
    }
}

class Dog extends Animal {
    @Override
    void sound() {
        super.sound();
        System.out.println("Dog Barks");
    }
}

public class Main {
    public static void main(String[] args) {
        Dog d = new Dog();
        d.sound();
    }
}
```



Animal Sound
Dog Barks